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Factors affecting generative artificial intelligence, such as ChatGPT, use in higher education: An application of technology acceptance model

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Abstract

The adoption of generative artificial intelligence (GAI) tools, such as ChatGPT, in higher education presents numerous opportunities and challenges. The use of GAI technologies in various fields, including education, has accelerated as technology develops. The widely used language model ChatGPT, developed by OpenAI, has become progressively more important, especially in the field of education. This study employs the technology acceptance model to investigate the factors influencing the employment of ChatGPT within the higher education sector of Pakistan. This study employed the PLS-SEM method for probing data collected from 368 Pakistani university students. The findings indicate that ChatGPT trust positively mediates the affiliation between ChatGPT self-efficacy, ChatGPT actual use, ChatGPT use for information and ChatGPT use for interaction. Further, ChatGPT usefulness and ChatGPT ease of use significantly moderate the association between ChatGPT self-efficacy and ChatGPT trust. Educators must encourage students to use ChatGPT safely to preserve their critical thinking, problem-solving abilities and creativity during assessments. This study contributes to understanding generative AI tools such as ChatGPT that are used in educational settings and provides insights for administrators and policymakers aiming to implement these technologies effectively.

KEYWORDS
ChatGPT actual use, ChatGPT ease of use, ChatGPT self-efficacy, ChatGPT trust, ChatGPT usefulness, generative artificial intelligence

Key insights

What is the main issue that the paper addresses?

The main issue addressed by this paper is the influence of ChatGPT self-efficacy on actual use, information and interaction through trust, ease of use and usefulness of generative AI technology in Pakistan's higher education sector. This study also helps organisations and individuals promote online education technologies such as generative AI and improve their educational goals.

What are the main insights that the paper provides?

The paper's main insight is that generative AI tools like ChatGPT have created new avenues for the education sector. This paper emphasises encouraging safe ChatGPT use to maintain students' critical thinking, problem-solving and creativity. It offers educators, administrators and policymakers valuable insights into integrating generative AI tools like ChatGPT into educational environments.

INTRODUCTION

Artificial intelligence (AI) in higher education has opened new paths for teaching and learning, with ChatGPT emerging as a prominent tool in this digital transformation (Maheshwari, 2024). AI has the power to automate jobs and tasks, handle massive volumes of data and offer predictive insights that transform a variety of daily activities (Habibi et al., 2023). According to Paul et al. (2023), the academic scene has undergone substantial changes due to generative AI incorporating ChatGPT (Generative Pre-trained Transformer), developed by OpenAI. AI has advanced significantly from its 1956 Dartmouth conference introduction, especially in the field of language creation, and is now represented by transformer-based models like ChatGPT (Zhang, Zhu & Su, 2023). Although there are many advantages to adopting ChatGPT in higher education (HE), there are drawbacks as well, such as the dissemination of false material and worries about data security and privacy. Additionally, by producing drafts of research papers, summarising scientific books and making it easier to retrieve materials, ChatGPT speeds up the research process as a research tool (Menon & Shilpa, 2023).

The effectiveness of e-learning programmes relies heavily on user acceptance and their willingness to integrate digital tools into their learning experience (Xu et al., 2024). Particularly, students' openness to using these tools for learning and interaction is crucial (Shahzad, Xu & Javed, 2024). Over the past two decades, research (Lim et al., 2023) has extensively explored individuals' usage of GAI technologies (ChatGPT) in HE. According to the technology acceptance model (TAM) put forward by Davis et al. (1989), perceived usefulness (PU) and perceived ease of use (PE) are key determinants of users' adoption of technology (Bernabei et al., 2023). PE is the extent to which users feel that using ChatGPT is simple, whereas PU is the extent to which users see benefits and value from using it

(Saif et al., 2023). Previous studies have continuously proven the significance of these two aspects in moulding users' behavioural intentions (Al-Abdullatif, 2023; Shahzad, Xu & Zahid, 2024) on several technologies, including AI chatbots and ChatGPT. Students, driven by ChatGPT self-efficacy, trust, ChatGPT usefulness and ChatGPT ease of use, exhibit an increased inclination towards utilising modern technologies such as ChatGPT. They thought of these new methods as opportunities to enhance their skills, facilitate learning and provide flexibility, further motivating their adoption (Xu et al., 2024).

The idea is self-efficacy, which is defined as a person's confidence in their own capacity to perform or acquire particular activities. ChatGPT in academia may reduce teaching and learning responsibilities, give insights into students' learning progress and promote classroom innovation by simplifying grading, supervision and feedback (Zhang, Maeda, et al., 2023). Nonetheless, the impact of ChatGPT self-efficacy has not been thoroughly explored in empirical researches (Bubou & Job, 2022; Mlambo et al., 2020), necessitating this study's investigation. Students' accreditation of ChatGPT trust reflects how reliable, respectable and trustworthy they believe AI technologies like ChatGPT to be. ChatGPT adoption requires people to feel secure and confident in the technology's capabilities and ethical application, making trust essential. A high level of technological confidence facilitates the use of generative AI tools. It is often acknowledged that trust is important for technological adoption (Ali et al., 2023). A past study (Hyun Baek & Kim, 2023) explained how trust perceives the credibility and reliability of ChatGPT use by students. Trust is essential in online purchase decision-making and influences students' interactions with AI-powered tools like ChatGPT.

Students' degree of trust is raised by ChatGPT self-efficacy, but positive usability enhances the bond even more. ChatGPT can help students find and understand information quickly. By interacting with ChatGPT, students can practice various skills, including writing, argumentation and critical thinking (Shahzad, Xu & Javed, 2024). Their self-efficacy in these areas can increase as they receive feedback and see improvements (Xu et al., 2024). The moderating role of ChatGPT usefulness and ease of use between ChatGPT self-efficacy and ChatGPT trust has been considered occasionally. Consequently, the study's goal is to examine ChatGPT self-efficacy's impact on ChatGPT's actual use, information and interaction through ChatGPT trust in Pakistan's HE sector. It also enlightens the moderating role of ChatGPT usefulness and ChatGPT ease of use in relation to ChatGPT self-efficacy and ChatGPT trust in Pakistan's HE sector. The study presents students' willingness to utilise ChatGPT for learning and knowledge-sharing objectives, which benefit Pakistan's educational system. By considering user concerns, it also assists institutions in promoting online education technologies and enhancing their security.

There are studies (Bilquise et al., 2023; Shahzad, Xu & Javed, 2024) on students' intent to use AI platforms for learning. Nevertheless, there is a clear research gap regarding students' self-efficacy of ChatGPT, especially when it comes to their aspirations to adopt and use it in higher education. Previous research has concentrated on several adoption factors; however, no one has utilised ChatGPT due to the aspect of self-efficacy and trust. Notably, a specific empirical study on ChatGPT that was carried out in Poland (Strzelecki, 2023) is noteworthy because it looks into students' acceptance and practical use of ChatGPT. Very rare studies have been conducted on this specific topic of inquiry; the majority of ChatGPT research has consisted of editorials, reviews, comments and general discussions. The current study focuses on ease of use and usefulness, acceptance and adoption of ChatGPT among Pakistani university students to close the research gap in the ChatGPT literature by examining all relevant elements. Individuals utilise ChatGPT to develop effective instruction, assessment and support initiatives and increase students' responsiveness to complete educational tasks.

THEORETICAL UNDERPINNING AND HYPOTHESIS DEVELOPMENT

Technology acceptance model

The study has chosen the TAM as the essential foundation for ChatGPT adoption, an AI tool that promotes self-regulated learning. The selection of the TAM for this study was made for several reasons (Davis, 1989). Since the TAM was initially created to predict and elucidate students' acceptance and usage of ChatGPT, it is first and foremost the best tool for analysing ChatGPT adoption and use within the context of ChatGPT ease of use (CEOU), ChatGPT usefulness (CU) and ChatGPT trust (CTR). In the context of ChatGPT, usefulness is the degree to which a user believes that using ChatGPT will enhance their performance or productivity (Hyun Baek & Kim, 2023). For instance, if users perceive ChatGPT as a useful tool for obtaining information, completing tasks or assisting in decision-making, they are more likely to use it. Users with high self-efficacy regarding their ability to interact with AI-driven tools like ChatGPT might believe more strongly in the utility of ChatGPT to effectively aid their tasks (Niu & Mvondo, 2024). With ChatGPT, perceived ease of use refers to the extent to which a user believes that using ChatGPT will be free of effort. This includes how user-friendly and accessible the interface and interaction are. Individuals who are confident in their technical skills are likely to find ChatGPT easier to use. This comfort can increase their likelihood of using the technology, as the ease of interaction aligns with their skills (Ali et al., 2023). Trust involves the belief that the system is reliable, secure and operates with integrity. If users trust ChatGPT to provide accurate and unbiased information, their perceived usefulness of the platform increases. Users who trust their judgement in using complex systems are more likely to trust the outputs from such systems, linking self-efficacy with a higher level of trust in the technology (Lai et al., 2023).

Moreover, the TAM offers the adaptability to modify and alter its components to suit certain settings and technological advancements. In order to meet the unique requirements and features of integrating ChatGPT into HE, the model must be adaptable (Davis, 1989). Our study builds upon the frameworks of the TAM to explore the impact of CEOU, CU and CTR on the actual usage of ChatGPT, particularly focusing on information acquisition and interactive engagement within the education sector. By leveraging the TAM, we aim to provide insights into how students' confidence in their technological abilities influences their adoption and utilisation of ChatGPT for educational purposes. This approach allows us to delve into the nuanced dynamics of technology acceptance and usage within the academic context, shedding light on the factors that shape students' behaviours and intentions regarding ChatGPT integration in learning environments (Al-Adwan et al., 2023).

ChatGPT self-efficacy

The concept of self-efficacy, which refers to the conviction that one can achieve in particular circumstances, holds significant relevance for investigating the effects of AI on learners (Liu et al., 2024). Self-efficacy can influence a student's approach to problem-solving and learning, affecting both the process and the outcomes of their educational experiences (Xu et al., 2024). ChatGPT self-efficacy can serve as a supplementary educational tool, providing explanations, aiding in comprehending complex subjects and offering instant feedback on assignments. This support can enhance students' understanding of course material, potentially increasing their academic self-efficacy by giving them a sense of mastery over challenging topics

(Bin-Nashwan et al., 2023). The use of generative AI tools raises concerns about academic integrity. The ability of ChatGPT to generate high-quality content with minimal input can lead to issues like plagiarism and a diminished emphasis on developing critical thinking and writing skills. Universities must address these challenges by creating guidelines that define ethical AI use, ensuring that students enhance rather than bypass their learning (Zhang, Maeda, et al., 2023). Engaging with AI tools like ChatGPT can influence students' self-efficacy differently. The ease and efficiency of obtaining information and creating content can boost confidence in their academic abilities (Xu et al., 2024). Generative AI tools in education also highlight the need for developing future-oriented skills that AI cannot easily replicate, such as critical thinking, creativity and interpersonal skills. By emphasising these areas, educational institutions can help students enhance their employability and adapt to a rapidly changing job market influenced by AI technologies (Alqahtani et al., 2023). ChatGPT self-efficacy in university students enhances educational experiences and potential risks that must be managed. By fostering an environment that encourages the ethical use of AI and focuses on developing essential human skills, universities can ensure that students thrive academically and are well prepared for the future (Davis, 1989).

Mediating role of ChatGPT trust

ChatGPT trust is pivotal in integrating generative AI technologies into the academic pursuits of Pakistani university students (Bilquise et al., 2023). Trust in ChatGPT reflects students' confidence and reliance on this AI tool's credibility and effectiveness in assisting their learning endeavours (Hyun Baek & Kim, 2023). It encompasses factors such as reliability, accuracy and user satisfaction, which collectively influence students' attitudes and behaviours towards utilising ChatGPT for educational purposes. Furthermore, ChatGPT trust is a crucial intermediary between ChatGPT self-efficacy and various dimensions of ChatGPT usage among Pakistani university students (Ali et al., 2023). Firstly, ChatGPT trust mediates the connection among ChatGPT self-efficacy and the actual utilisation of ChatGPT. Students with higher levels of self-efficacy in using ChatGPT are more likely to trust its capabilities, increasing their propensity to engage with the tool in practical learning tasks and activities (Niu & Mvondo, 2024). Secondly, ChatGPT trust association leads to acquiring information. When students feel confident in their ability to effectively utilise ChatGPT, their trust in its capacity to deliver accurate and relevant information is strengthened, leading to increased reliance on the tool for informational purposes (Paul et al., 2023). In a past study, ChatGPT trust served as a mediator and explained how individuals use ChatGPT for interactive purposes. Students with higher levels of self-efficacy are more inclined to trust ChatGPT's ability to facilitate meaningful interactions and engagements, thereby fostering greater utilisation of the tool for collaborative learning and communication within academic settings (Ma & Huo, 2023). Overall, these mediation pathways underscore the integral role of trust in shaping students' utilisation of ChatGPT for diverse educational activities. Based on the above arguments, we hypothesise that:

> **H1.** ChatGPT trust acts as a mediator between ChatGPT self-efficacy and the actual use of ChatGPT among Pakistani university students.

> **H2.** ChatGPT trust mediates the association between ChatGPT self-efficacy and ChatGPT use for information by Pakistani university students.

> **H3.** ChatGPT trust mediates the connection among ChatGPT self-efficacy and ChatGPT use for interaction among Pakistani university students.

Moderating role of ChatGPT usefulness

Usefulness, in this context, is how valuable or beneficial users find ChatGPT for achieving their goals. When ChatGPT is perceived as highly useful, even users with moderate self-efficacy might develop higher trust in the tool (Shahzad, Xu & Javed, 2024). Conversely, if ChatGPT is perceived as less useful, high self-efficacy might not necessarily translate into high trust. In our study focusing on ChatGPT application by university students in HE, ChatGPT usefulness denotes how individuals perceive ChatGPT as valuable and effective in facilitating their academic tasks and objectives (Niu & Mvondo, 2024). Within this context, the utility of ChatGPT plays a crucial role in influencing its acceptance and adoption among students. Empirical evidence consistently supports a positive relationship between ChatGPT usefulness and the acceptance of technological innovations within educational settings (Adarkwah et al., 2018). For instance, ChatGPT's usefulness has been shown to predict students' willingness to utilise it for tasks such as information retrieval, problem-solving and academic interaction. Moreover, ChatGPT's usefulness is a significant factor in overcoming potential barriers to its adoption and sustained use within educational environments (Ayinde et al., 2023). Within HE, ChatGPT's usefulness influences students' attitudes and intentions towards utilising it to enhance their learning experiences. ChatGPT's usefulness also serves as a significant determinant of students' trust in its capabilities, fostering confidence in its ability to effectively support their academic endeavours (Lai et al., 2023). When ChatGPT usefulness is higher, the effect of ChatGPT self-efficacy on ChatGPT trust becomes more pronounced, highlighting the crucial role of ChatGPT usefulness in shaping students' trust in the system within HE (Hyun Baek & Kim, 2023). Therefore, we hypothesise that:

> **H4.** ChatGPT usefulness moderates the affiliation between ChatGPT self-efficacy and ChatGPT trust.

Moderating role of ChatGPT ease of use

Ease of use, within the framework of the TAM, holds particular significance for students utilising ChatGPT within HE contexts (Al-Abdullatif, 2023). It represents the extent to which students perceive ChatGPT as effortless and intuitive to use in their academic endeavours. This aspect is critical in shaping students' intentions and attitudes towards incorporating ChatGPT into their learning (Bilquise et al., 2023). Empirical studies (Davis, 1989; Hyun Baek & Kim, 2023) have consistently demonstrated that ChatGPT's ease of use influences students' willingness to adopt various technological innovations, such as online learning platforms and educational tools. Moreover, ease of use significantly impacts students' acceptance of GAI, including ChatGPT, within the educational domain. In the education sector, ChatGPT ease of use is essential for fostering positive intentions towards utilising ChatGPT for academic purposes (Hyun Baek & Kim, 2023). Students need to perceive ChatGPT as user-friendly and effortless to navigate to fully engage with its functionalities for tasks such as information retrieval, problem-solving and collaborative learning. Ease of use is imperative in shaping students' trust in ChatGPT capabilities and reliability. When students find ChatGPT easy to use and navigate, they are more likely to trust its recommendations and rely on it as a valuable resource for their academic pursuits (Bernabei et al., 2023). Past research (Maheshwari, 2024) has highlighted the importance of offering easy-to-use and intuitive educational technologies to enhance students' learning experiences in HE. When ChatGPT is perceived as easy to use, the positive association between self-efficacy and trust is likely to be stronger. If users believe they can use ChatGPT effectively (high self-efficacy) and find it easy to use, they are more likely to trust the tool (Shahzad, Xu & Javed, 2024). Ensuring that ChatGPT is perceived as user-friendly and effortless enhances students' trust in higher education (Ali et al., 2023). Based on the following assertions, we hypothesise that:

H5. ChatGPT ease of use moderates the association among ChatGPT self-efficacy and ChatGPT trust.

The theoretical model of the research is described in Figure 1.

RESEARCH METHODOLOGY

This study examined learners' tendency to use ChatGPT to advance the longevity of HE. The study centred on students actively involved in blended learning and utilising AI technologies in their educational endeavours. The poll included demographic questions that participants answered on a five-point Likert scale, where 5 denotes 'strongly agree' and 1 denotes 'strongly disagree'. The surveys were distributed and self-administered, and respondents were requested to share insights about ChatGPT self-efficacy, trust and its impact on their educational experiences, such as ChatGPT use. The data analysis employed the partial least squares structural equation modelling (PLS-SEM) technique implemented by SmartPLS 4 software.

Data collection procedure

The study included students who were currently enrolled in traditional institutions in Pakistan, with a specific emphasis on promoting and integrating the use of ChatGPT in the country's

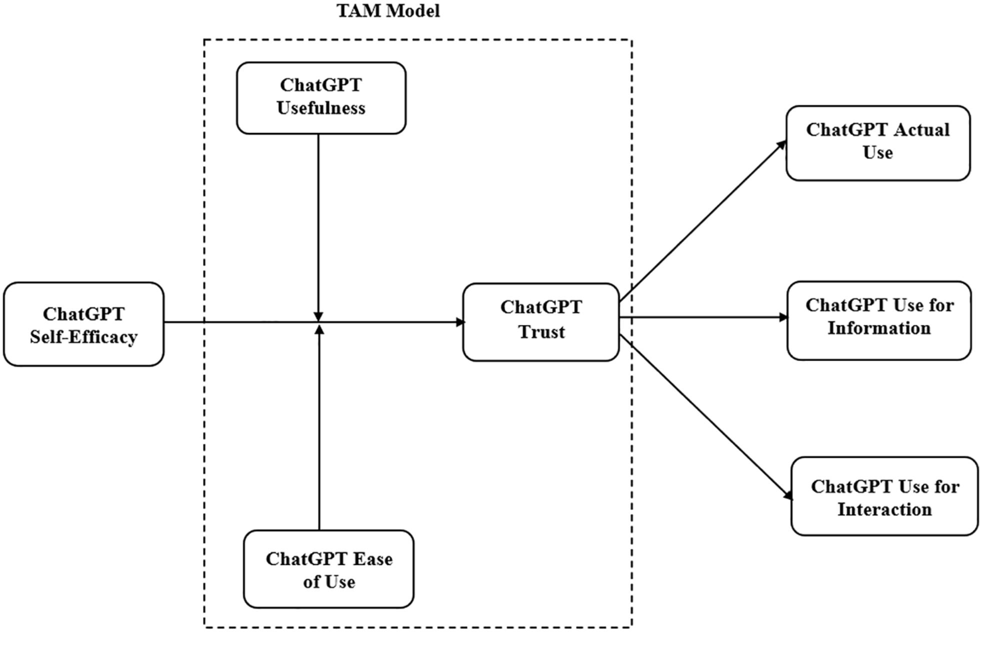

FIGURE 1 Theoretical model.

field of education. ChatGPT has become a popular educational tool for university students on many campuses (Ma & Huo, 2023). The data-gathering process consisted of randomly selecting six esteemed universities in Punjab, Pakistan, which were determined based on rankings provided by the Higher Education Commission (HEC) of Pakistan. These universities were selected based on their curriculum, educational objectives and faculty proficiency, as recognised by the HEC. The individuals who met the criteria to participate in this survey were considered as the target population (Lai et al., 2023). This extensive research gathered a representative sample from the intended demographic to obtain crucial data. Convenience sampling was used because there was no pre-existing sampling frame available, hence non-probability sampling was adopted. The sample size was determined using a strategy recommended by Roscoe et al. (1975), which states that more than 30 but fewer than 500 participants should be included in an empirical study. The researchers carried information about the study's variables from participants through multiple means, such as emails, WhatsApp and social media posts.

We sent an invitation to students to willingly participate in the completion of the questionnaire. The main method used to collect data for this survey was self-completed questionnaires. The use of self-administered questionnaires expedited data gathering from a larger sample size. Questionnaires were manually and electronically given to eligible respondents to increase the response rate. Additionally, the research team requested permission from the relevant campuses to gather study data and convey its aims. A total of 510 questionnaires were issued to university students, and 470 participants responded. Nevertheless, the data was further examined due to the discovery of multiple incomplete surveys. Several questions were excluded from the analysis because they had errors, biases or missing responses. After conducting a comprehensive examination of the data, we were left with a total of 368 valid cases used for study analysis. Therefore, the ultimate rate of response calculated using the data that may be effectively utilised is 72%. This sample size conforms to guidelines that suggest empirical research (Roscoe et al., 1975) should include more than 30 but less than 500 people. However, the sample data includes 224 male respondents (61%) and 144 female respondents (39%). Table 1 presents further information about the respondents.

Instrumentations

The questionnaire is divided into two sections. The preliminary section collects demographic material from participants, encompassing factors such as gender, age and education. The second section of the questionnaire was customised to assess the specific variables relevant to this inquiry. The questions were derived from previous study sources. Participants' responses were gathered using a five-point Likert scale. ChatGPT self-efficacy (CSE) in education refers to an individual's confidence in completing tasks connected to learning and academic success. The seven items of CSE were adopted from a past study (Compeau & Higgins, 2017). Trust in ChatGPT can be attributed to its precision in delivering pertinent and beneficial answers and meeting user expectations. ChatGPT trust (CTR) was measured using six items taken from Tarhini et al. (2017). The utility of ChatGPT in education stems from its ability to produce customised learning resources and boost student involvement in learning encounters. ChatGPT usefulness (CU) was evaluated using five items taken from Davis (1989). ChatGPT's ease of use in education enables students to create content and effortlessly obtain immediate responses to inquiries. ChatGPT ease of use (CEOU) was assessed using six items taken from Davis (1989). Actual use of ChatGPT includes faith in one's capacity to set and attain educational objectives and persevere against hindrances. Five items of actual use

TABLE 1 Participant information.

Demographics	Distribution	*n* = 368
Gender	Male	224 (61%)
	Female	144 (39%)
Age	20–25 years	108 (29%)
	26–33 years	149 (41%)
	Above than 3 years	111 (30%)
Education	Bachelor	124 (34%)
	Master's	151 (41%)
	PhD	93 (25%)
Specialisation	Management sciences	119 (32%)
	Business administration	98 (27%)
	Hotel management	106 (29%)
	Others	45 (12%)
ChatGPT experience in education	Average	38 (10%)
	Good	148 (40%)
	High	182 (50%)

of ChatGPT (CAU) were suggested by Faqih and Jaradat (2021). The desire to employ ChatGPT for information mentions an individual's purpose and intent to incorporate these tools into their education to enhance the usage of information. ChatGPT use for information (CUI) was assessed using six items obtained from Kim et al. (2019). Interaction with ChatGPT impacts the level of involvement individuals have in learning activities, the approaches they choose to handle academic tasks and their overall academic achievement. The construct of ChatGPT use for interaction (CUIT) was evaluated using six items obtained from Wang et al. (2022). Table 2 shows the item measurements in detail.

Pilot study

Additionally, a pilot study was conducted to evaluate the questionnaire's quality: 42 university students with knowledge of ChatGPT use in education participated. Based on the findings, the researcher revised the item's phrasing and a second round of pilot testing was conducted. Following the noteworthy results of the pilot study, the instrument was fully prepared for data collection. Another reason to conduct a pilot study was to verify that the respondents accurately understood each question item and that the questions were thorough and easy to understand. A convenience sampling strategy that is non-probability was employed in this investigation. It is unknown how likely it is that any given case will be selected from the total population in non-probability samples (Shahzad, Xu & Baheer, 2024). The easiest instances to obtain for the sample are chosen at random in convenience sampling; this process continues until the necessary example size is reached.

ANALYSIS AND RESULTS

The data was analysed using the PLS-SEM technique, which is a regularly used way to investigate correlations between several construct variables. The choice of PLS-SEM

TABLE 2 Factors analysis.

Constructs	Items		FL	α	CR	AVE	Source
ChatGPT self-efficacy (CSE)				0.926	0.941	0.696	Compeau and Higgins (2017)
	CSE1	I can use ChatGPT, which I can access, to accomplish my academic objectives	0.886				
	CSE2	I can use ChatGPT tools and my own personal effort to learn	0.895				
	CSE3	I'm inspired to study ChatGPT-based technologies	0.774				
	CSE4	Using ChatGPT for instructional purposes has become commonplace	0.801				
	CSE5	I have to research and rewrite my lecture using ChatGPT	0.916				
	CSE6	I believe ChatGPT technology is now widely used	0.792				
	CSE7	Applications for ChatGPT should be worth more to me than the desired sacrifice	0.761				
ChatGPT trust (CTR)				0.966	0.973	0.855	Tarhini et al. (2017)
	CTR1	I believe utilising ChatGPT for communication would be safe enough	0.892				
	CTR2	I have faith that whatever I do on ChatGPT will be private and safe	0.916				
	CTR3	I believe ChatGPT would protect the confidentiality of my personal information	0.933				
	CTR4	I believe ChatGPT will keep my personal information safe from prying eyes	0.927				
	CTR5	I'm hoping that ChatGPT will now have a secure atmosphere	0.941				
	CTR6	I believe ChatGPT will be useful for organising my studies and courses	0.939				

TABLE 2 (Continued)

Constructs	Items		FL	α	CR	AVE	Source
ChatGPT usefulness (CU)				0.926	0.945	0.775	Davis (1989)
	CU1	I receive support from ChatGPT with all of my academic tasks	0.917				
	CU2	I believe employing ChatGPT would help me learn more	0.849				
	CU3	ChatGPT would provide me with answers to all of my inquiries and expectations	0.887				
	CU4	I could improve the effectiveness and quality of my learning with ChatGPT	0.940				
	CU5	No matter where I am, ChatGPT will provide me with instant access to study-related issues	0.802				
ChatGPT ease of use (CEOU)				0.884	0.907	0.621	Davis (1989)
	CEOU1	ChatGPT would make it simple to use and modify	0.760				
	CEOU2	It would be simple for me to access ChatGPT during my studies	0.791				
	CEOU3	ChatGPT would be simple to use and comprehend	0.854				
	CEOU4	I believe that using ChatGPT only requires rudimentary abilities	0.805				
	CEOU5	It would be easy to obtain study-related information through ChatGPT	0.726				
	CEOU6	I use ChatGPT because it makes it easier to find answers to inquiries and finish activities	0.786				

(Continues)

TABLE 2 (Continued)

Constructs	Items		FL	α	CR	AVE	Source
ChatGPT actual use (CAU)				0.856	0.897	0.635	Faqih and Jaradat (2021)
	CAU1	I believe that ChatGPT is essential to my learning at all times	0.737				
	CAU2	In order to learn somewhere, I believe I have to use ChatGPT	0.813				
	CAU3	Using ICT technology, I consider, will help me to keep going	0.791				
	CAU4	In the future, I plan to keep utilising ChatGPT	0.823				
	CAU5	I will make an effort to use ChatGPT for my education	0.818				
ChatGPT use for information (CUI)				0.900	0.923	0.666	Kim et al. (2019)
	CUI1	I will consider using ChatGPT for my subject information	0.860				
	CUI2	I am planning to use ChatGPT to get novel information	0.871				
	CUI3	I will continue to use ChatGPT in the future	0.844				
	CUI4	I will inform others regarding the goodness of ChatGPT	0.772				
	CUI5	I will use the resources essential to use ChatGPT	0.775				
	CUI6	I pay reasonable attention to using ChatGPT for my education	0.765				

TABLE 2 (Continued)

Constructs	Items		FL	α	CR	AVE	Source
ChatGPT use for interaction (CUIT)				0.919	0.937	0.714	Wang et al. (2022)
	CUIT1	ChatGPT, when used for educational purposes, has become customary	0.875				
	CUIT2	I have used ChatGPT to study and revise my lesson	0.876				
	CUIT3	It has become commonplace to use ChatGPT	0.854				
	CUIT4	It's easy to use ChatGPT technology, and I can find out more	0.786				
	CUIT5	It's easy to learn about technological innovations like ChatGPT	0.865				
	CUIT6	Using ChatGPT is easy and clear to grasp	0.809				

Abbreviations: AVE, average variance extracted; CR, composite reliability; FL, factor loadings; VIF, variance inflation factor; α, Cronbach's alpha.

was made based on its flexibility in handling small sample numbers and its independence from the assumption of normal data distribution (Hair et al., 2020). The statistical software SMARTPLS 4 enabled the analysis of data and the testing of models. SMARTPLS 4 software is frequently applied in studies to examine intricate connections between latent and observable variables. The preliminary hypothesis was tested using models of structural assessment and measurement, as suggested by Ma and Huo (2023). Initially, construct reliability and convergent and discriminant validity were suggested by the measurement models. The second approach employed a structural model that comprehensively assesses the hypothesis's relevance. It promotes the efficacy, ease and accuracy of statistical analysis. PLS-SEM is the most appropriate analysis method when it comes to developing new, theoretically complicated models that achieve objectives (Strzelecki, 2023).

Common bias method

As the data was gathered via own administration and a mail survey, it was not possible to evaluate non-response bias: all participants completed the questionnaires within the allotted time limit, returned their responses within that timeframe and it was impossible to examine differences between non-respondents and respondents (Podsakoff et al., 2003). PLS-SEM is used in this work to demonstrate CMB. This work identified common approach bias by using a variance inflation factor (VIF) generated through thorough collinearity testing, as suggested by Shahzad, Xu and Javed (2024). The analysis's findings indicated that there was no CMB contamination because each construct's VIF value was less than 3.30. Second, Harman's one-factor test was utilised, along with SPSS software, to fully evaluate the potential for CBM in this study. As per Podsakoff et al. (2003), the usual common method bias criterion of 50% was significantly exceeded by our analysis, which found that a single component only contributed 22% of the variation. The aforementioned results strongly suggest that common technique bias does not compromise the integrity of this study.

Measurement model

Prior to using the data for preliminary hypothesis testing, the measurement model of the SEM technique needed to be assessed. Validity and construct reliability are the two procedures used to select the validation criteria in the measurement model procedure (Hair et al., 2020). Cronbach's alpha, composite reliability (CR) and average value extracted (AVE) are the assessment benchmarks in the construct reliability approach. According to past research, data has considerable internal consistency reliability when the Cronbach's alpha and CR values are higher than 0.7. Constructs with valid reliability are indicated by AVE values greater than 0.5. Factor loadings also need to be greater than 0.7 for the observable variables to adequately explain the latent variables (Hair et al., 2014). As supported by Sarstedt et al. (2014), factor loadings were utilised to demonstrate construct validity and dependability for the model's acceptable degree of fit. The Cronbach's alpha values were as follows: CSE, 0.926; CTR, 0.966; CU, 0.926; CEOU, 0.884; CAU, 0.856; CUI, 0.900; CUIT, 0.919. CR values: CSE, 0.941; CTR, 0.973; CU, 0.945; CEOU, 0.907; CAU, 0.897; CUI, 0.923; CUIT, 0.937. AVE values: CSE, 0.696; CTR, 0.855; CU, 0.775; CEOU, 0.621; CAU, 0.635; CUI, 0.666; CUIT, 0.714. The measurement model therefore has good convergent validity for the data. Table 2 displays the factor loading values for university student measurement models.

First, the discriminant validity was assessed via the application of the Fornell–Larcker method. The square root of the AVE, represented by the diagonal numbers in bold in Table 3, ought to be higher than the correlation coefficient between each variable (Fornell &

TABLE 3 Discriminant validity (Fornell–Larcker criteria).

Constructs	1	2	3	4	5	6	7
1. CAU	**0.797**						
2. CEOU	0.492	**0.788**					
3. CSE	0.435	0.615	**0.834**				
4. CTR	0.461	0.702	0.729	**0.925**			
5. CU	0.368	0.570	0.827	0.709	**0.880**		
6. CUI	0.457	0.551	0.518	0.540	0.591	**0.816**	
7. CUIT	0.755	0.601	0.464	0.538	0.384	0.643	**0.845**

Abbreviations: CAU, ChatGPT actual use; CEOU, ChatGPT ease of use; CSE, ChatGPT self-efficacy; CTR, ChatGPT trust; CU, ChatGPT usefulness; CUI, ChatGPT use for information; CUIT, ChatGPT use for interaction.

Larcker, 1981). In this study, the correlation value among the latent variables is less than the AVE value for data collected from university students.

Second, HTMT makes a more explicit distinction between constructs than the Fornell–Larcker criteria; it is thought to be a better approach (Hair et al., 2014). All of the HTMT values in Table 4 fall within the permissible range and are deemed to have sufficient discriminant validity. The discriminant validity can thus adequately explain the suggested model, which is sufficient.

Third, in order to prevent items from cross-loading onto other factors, it is necessary to analyse the pattern of factor loadings and make sure that items primarily load on the factors they are supposed to load on (Henseler et al., 2015). All cross-loading values within the range are displayed in Table 5.

Structural model

In the second step of assessing PLS-SEM, bootstrapping was employed to determine the importance of the structural route. In the bootstrapping method, 5000 subsamples were evaluated using replacements to check for mistakes, and the outcomes guided the projected *t*-values for the significance test of the proposed model (Sarstedt et al., 2014). The significance of the connection can be determined by taking into account the *t*-value, as per Hair et al. (2020). A significant relationship is shown in this study with a *t*-value greater than 1.96 and a *p*-value less than 0.05. Even though PLS-SEM is frequently thought to be resistant to deviations from normalcy assumptions, comprehension of the data distribution might be helpful in evaluating outcomes and guaranteeing the validity of the results (Hair et al., 2014). The bootstrapping methods approach to data normalcy for structural models is shown in Table 9. Multicollinearity in the study model is assessed in the first phase using the VIF for each construct (Saif et al., 2023). To ensure that there is no multicollinearity problem with the constructions, the VIF value should not be greater than 5.

Much statistical data is provided by the structural assessment model, for example pathway coefficient values, impact scale (f^2), *t*-values, predictive relevance (Q^2) and coefficient of determination (R^2) (Hair et al., 2020). According to Sarstedt et al. (2014), the R^2 value indicates the percentage of a dependent variable's change that is brought about by an independent variable or the other variables in a regression model. Based on the data, the model used in this study is sound, as shown in Figure 2, where the R^2 value is high. The findings show that the independent constructs' R^2 values, like CTR, account for 21%, 29% and 29% of the variation in CAU, CUI and CUIT, respectively. Table 6's range is where the R^2, Q^2 and f^2 values are located.

TABLE 4 Discriminant validity (HTMT criteria).

Constructs	1	2	3	4	5	6	7
1. CAU							
2. CEOU	0.553						
3. CSE	0.485	0.606					
4. CTR	0.500	0.784	0.767				
5. CU	0.411	0.559	0.893	0.748			
6. CUI	0.529	0.598	0.561	0.572	0.636		
7. CUIT	0.848	0.654	0.501	0.566	0.416	0.728	

Using the standardised root mean square residual (SRMSR) values for the constructs, the goodness-of-fit (GoF) index was also calculated (Gefen & Straub, 2005). The model fits the data well, as indicated by the value of the SRMR (0.115). Other model fitness indicators (e.g., $\chi^2 = 13626.911$ and NFI = 0.472) are shown in Table 7.

Importance–performance map analysis (IPMA) explains the contrast of importance as follows: a one point increase in the predecessor construct will increase the performance of the target performance by the total effect size (i.e., importance) of the same predecessor construct (Sarstedt et al., 2024). A list of importance–performance values is provided in Table 8.

The final assessment of hypothesis development is revealed in Table 9. The results of this study, which examined the relationship between CSE and ChatGPT trust, showed a significant correlation ($\beta = 0.165$, $t = 2.141$, $p < 0.05$). The relationship between ChatGPT trust and CAU, CUI and CUIT was investigated. A significant correlation was found in the results ($\beta = 0.461$, 0.540, 0.538; $t = 7.656$, 11.180, 10.886; $p < 0.05$). This study looked at the relationship between CU, CEOU and CTR. The data showed a significant correlation ($\beta = 0.265$, 0.547; $t = 2.850$, 11.533; $p < 0.05$).

After the direct effect, this study tested the mediation effect. According to the findings of the mediation analysis, ChatGPT trust has a positive mediating effect on the link between CAU, CUI and CUIT ($\beta = 0.071$, 0.089, 0.087; $t = 3.161$, 2.176, 2.002; $p < 0.05$). Therefore, H1, H2 and H3 are accepted.

The moderator can affect (strengthen or weaken) the relationship between two variables, according to Shahzad, Xu and Zahid (2024). The current study determined the moderating factor of CU and CEOU between CSE and ChatGPT trust. The findings of the moderation indicated that CU positively moderates the association between CSE and ChatGPT trust ($\beta = 0.133$, $t = 3.397$, $p < 0.05$). Therefore, we accept H4. Moving forward, CEOU acts as a significant moderator among CSE and ChatGPT trust ($\beta = 0.165$, $t = 3.650$, $p < 0.05$). Therefore, we accept H5.

DISCUSSION AND CONCLUSION

Generative AI (ChatGPT) has grown widely in use in modern culture, leading to significant changes in communication, work relationships and educational methods (Farrukh et al., 2024). Incorporating this technology into education is a powerful catalyst for innovation, providing new avenues to enhance educational experiences. In Pakistan, higher education is a crucial need for technologies that promote innovation, guarantee fair access to quality education and train a workforce that can compete globally (Saif et al., 2023). Therefore, this study examines how the belief in one's ability to use ChatGPT effectively influences the intention to use ChatGPT for information and interaction, as well as its

TABLE 5 Discriminant validity (cross loadings).

Items	CAU	CEOU	CSE	CTR	CU	CUI	CUIT
CAU1	0.737	0.441	0.353	0.395	0.296	0.536	0.705
CAU2	0.813	0.373	0.356	0.380	0.333	0.342	0.485
CAU3	0.791	0.318	0.316	0.344	0.294	0.295	0.457
CAU4	0.823	0.417	0.356	0.369	0.274	0.318	0.680
CAU5	0.818	0.400	0.344	0.340	0.262	0.304	0.670
CEOU1	0.363	0.760	0.326	0.464	0.311	0.407	0.434
CEOU2	0.414	0.791	0.344	0.430	0.301	0.442	0.504
CEOU3	0.417	0.854	0.360	0.498	0.309	0.414	0.508
CEOU4	0.385	0.805	0.334	0.448	0.290	0.380	0.467
CEOU5	0.264	0.726	0.596	0.741	0.554	0.349	0.361
CEOU6	0.465	0.786	0.677	0.881	0.656	0.544	0.546
CSE1	0.352	0.479	0.886	0.591	0.710	0.403	0.342
CSE2	0.351	0.484	0.895	0.588	0.730	0.401	0.332
CSE3	0.320	0.422	0.774	0.497	0.669	0.432	0.371
CSE4	0.412	0.564	0.801	0.626	0.707	0.559	0.446
CSE5	0.392	0.548	0.916	0.657	0.774	0.415	0.382
CSE6	0.345	0.561	0.792	0.648	0.659	0.394	0.413
CSE7	0.352	0.505	0.761	0.616	0.572	0.419	0.409
CTR1	0.359	0.700	0.689	0.892	0.659	0.409	0.406
CTR2	0.379	0.720	0.668	0.916	0.625	0.469	0.458
CTR3	0.409	0.746	0.653	0.933	0.613	0.510	0.512
CTR4	0.388	0.750	0.685	0.927	0.680	0.467	0.440
CTR5	0.500	0.766	0.679	0.941	0.680	0.563	0.574
CTR6	0.501	0.765	0.674	0.939	0.674	0.559	0.572
CU1	0.317	0.501	0.775	0.621	0.917	0.473	0.292
CU2	0.313	0.444	0.678	0.568	0.849	0.522	0.337
CU3	0.378	0.538	0.702	0.654	0.887	0.645	0.402
CU4	0.331	0.524	0.784	0.652	0.940	0.482	0.314
CU5	0.276	0.496	0.697	0.616	0.802	0.473	0.344
CUI1	0.283	0.459	0.482	0.480	0.576	0.860	0.371
CUI2	0.293	0.468	0.489	0.488	0.580	0.871	0.393
CUI3	0.268	0.447	0.492	0.476	0.555	0.844	0.357
CUI4	0.492	0.453	0.364	0.406	0.392	0.772	0.724
CUI5	0.495	0.446	0.341	0.394	0.372	0.775	0.721
CUI6	0.471	0.429	0.338	0.384	0.371	0.765	0.691
CUIT1	0.641	0.510	0.392	0.464	0.348	0.667	0.875
CUIT2	0.636	0.509	0.394	0.463	0.350	0.668	0.876
CUIT3	0.632	0.507	0.377	0.456	0.315	0.658	0.854
CUIT4	0.595	0.456	0.365	0.425	0.284	0.378	0.786
CUIT5	0.705	0.550	0.438	0.485	0.351	0.465	0.865
CUIT6	0.616	0.511	0.382	0.433	0.295	0.409	0.809

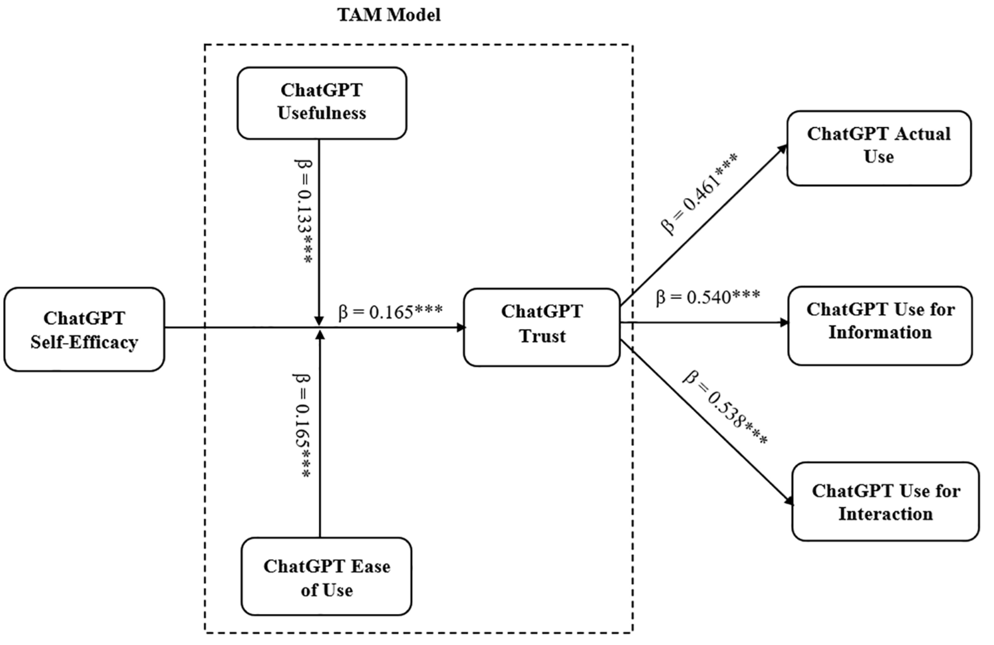

FIGURE 2 Structural model results (*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$).

TABLE 6 Structural model R^2, Q^2 and f^2.

Constructs	R^2	Q^2	f^2
CAU	0.213	0.130	0.189
CUI	0.292	0.192	0.412
CUIT	0.290	0.205	0.408
CTR	0.767	0.650	0.270
CU		0.654	0.090
CEOU		0.461	0.777
CSE		0.593	0.032

TABLE 7 Goodness of fit.

	Saturated model	Estimated model
SRMR	0.115	0.150
d_ULS	11.402	19.456
d_G	10.094	10.435
χ^2	13626.911	14012.535
NFI	0.472	0.458

actual usage, based on the level of trust in ChatGPT within Pakistan's educational sector. Furthermore, the study investigates how the usefulness and ease of use of ChatGPT affect the affiliation between an individual's self-efficacy in using ChatGPT and their level of faith in it. The study used a web-based questionnaire to assess the conceptual framework, drawing data from 368 students enrolled in Pakistani universities. The evaluation of this framework leads to the identification of five noteworthy research findings.

TABLE 8 Importance–performance map analysis for AU, CUI and CUIT.

Latent variables	Importance	Performance
CU	0.265	72.141
CEOU	0.547	65.141
CTR	0.540	72.754
CSE	0.165	72.354

TABLE 9 Hypothesis testing.

Relationships	β-values	SD	*t*-values	*p*-values	Outcome
Direct effects					
CSE → CTR	0.165	0.077	2.141	0.033	Yes
CTR → CAU	0.461	0.060	7.656	0.000	Yes
CTR → CUI	0.540	0.048	11.180	0.000	Yes
CTR → CUIT	0.538	0.049	10.886	0.000	Yes
CU → CTR	0.265	0.093	2.850	0.005	Yes
CEOU → CTR	0.547	0.047	11.533	0.000	Yes
Mediation effects					
CSE → CTR → CAU (H1)	0.071	0.023	3.161	0.002	Yes
CSE → CTR → CUI (H2)	0.089	0.041	2.176	0.030	Yes
CSE → CTR → CUIT (H3)	0.087	0.043	2.002	0.042	Yes
Moderation effects					
CSE*CU → CTR (H4)	0.133	0.039	3.397	0.001	Yes
CSE*CEOU → CTR (H5)	0.165	0.045	3.650	0.000	Yes

Abbreviations: CAU, ChatGPT actual use; CEOU, ChatGPT ease of use; CSE, ChatGPT self-efficacy; CTR, ChatGPT trust; CU, ChatGPT usefulness; CUI, ChatGPT use for information; CUIT, ChatGPT use for interaction; SD, standard deviation.

First, this study's results indicate that ChatGPT trust significantly mediates the connection between ChatGPT self-efficacy and ChatGPT actual use among Pakistani university students. Students' faith in ChatGPT use increases their belief in its reliability, use and effectiveness. A prior study (Ali et al., 2023) has shown that students in Pakistan's HE have different levels of trust in their abilities to use ChatGPT technology for educational objectives. People are ready to use ChatGPT technology for educational activities, such as gathering information and engaging in communication (Hyun Baek & Kim, 2023). This aligned with the results of a previous study (Xu et al., 2024), which confirms the claim that having greater ChatGPT self-efficacy might lead to a higher faith in technology. Increased ChatGPT trust, in turn, enables the effective incorporation of ChatGPT into educational activities. To establish confidence, it is important to make sure that technology is secure, reliable, easy to use and customised to suit the specific requirements of the educational setting (Abdelkader, 2023). Therefore, the results indicate that ChatGPT trust endorses the efficient use of ChatGPT in Pakistan's educational system.

Second, ChatGPT trust positively mediates the association between ChatGPT self-efficacy and ChatGPT use for information by Pakistani university students. ChatGPT trust acts as a connection to use ChatGPT and its efficacy for gathering educational information.

Students who firmly believe they are capable of using technology are more likely to have confidence in their ability to use ChatGPT efficiently (Ray, 2023). Individuals have confidence in their ability to effectively handle any obstacles or issues that may arise when using ChatGPT, which also includes faith in the technology. Pakistani educational institutions are encouraged to allocate resources towards training programmes that aim to boost the ChatGPT self-confidence of both teachers and students, enabling them to effectively employ it as a tool (Farrukh et al., 2024). People who have confidence are more likely to accept the information provided by ChatGPT technology, which in turn enhances the desire to use these tools to acquire information. ChatGPT trust includes issues about security and privacy, extending beyond the usefulness of ChatGPT technology (Paul et al., 2023). To cultivate confidence, institutions and technology providers must consider ensuring the dependability, safety and efficacy of ChatGPT tools. Open and clear communication about data management and privacy rules is crucial for developing trust (Shahzad, Xu & Javed, 2024).

Third, ChatGPT trust significantly mediates the connection between ChatGPT self-efficacy and ChatGPT use for interaction among Pakistani university students. Those who are confident in their capacity to use ChatGPT efficiently and engage in education may be more willing to use ChatGPT, if they have confidence in its dependability, safety and trustworthiness (Abdelkader, 2023). The importance of trust in ChatGPT arises from its ability to reduce perceived dangers and enhance the perceived advantages of using technology (de Andrés-Sánchez & Gené-Albesa, 2023). A past study (Hyun Baek & Kim, 2023) established that confidence in ChatGPT is a top priority for educational institutions and technology providers. This can be achieved by promoting ChatGPT interaction regarding security and privacy protocols, assuring unwavering dependability and providing thorough user education and assistance. On the other hand, ChatGPT trust has been thoroughly examined in earlier research on consumer behaviour, technology adoption and consumer psychology (Bilquise et al., 2023). Nonetheless, it is anticipated that students' perceptions of their own trustworthiness in ChatGPT performance and usage would mediate the platform's acceptance, greatly increasing the likelihood of ChatGPT adoption. Trust and efforts to improve it are important, and a more supportive environment can be created to successfully implement and use ChatGPT technologies in education (Haluza & Jungwirth, 2023).

Fourth, ChatGPT usefulness moderates the affiliation between ChatGPT self-efficacy and ChatGPT trust. ChatGPT's usefulness is highly valuable in multiple fields, especially for education, as it functions as a versatile instrument that enhances the quality of learning experiences (Niu & Mvondo, 2024). ChatGPT enables customised learning, encourages analytical thinking and enhances student involvement by generating individualised information, offering immediate feedback and promoting interactive discussions. When individuals trust their data is secure, they willingly use ChatGPT (Lai et al., 2023). Therefore, when people view technology as being secure, it positively impacts their confidence. Increased levels of ChatGPT self-efficacy can enable users to navigate and use these tools effectively, hence enhancing their sense of accomplishment and confidence (Bilquise et al., 2023). Generative AI provides users with the necessary information and skills to effectively use the technology. By cultivating users' confidence in their ability to utilise ChatGPT effectively, companies can enhance their impression of security and, as a result, cultivate trust in the technology (Strzelecki, 2023).

Fifth, ChatGPT ease of use moderates the association between ChatGPT self-efficacy and ChatGPT trust. ChatGPT ease of use usually involves ChatGPT intuitive functionality, easy-to-use interface and smooth interaction, which make it simple to create text and have conversations (Al-Abdullatif, 2023). ChatGPT ease of use acts as a bond between ChatGPT self-efficacy and ChatGPT trust. It can increase people's trust in digital learning tools or online educational platforms when parents and students with high technology self-efficacy communicate favourable experiences and suggestions to them (Hyun

Baek & Kim, 2023). According to Ali et al. (2023), a potential relationship exists between individuals' ChatGPT self-efficacy and their trust in digital learning tools, platforms and educational institutions, and ChatGPT ease of use in Pakistan's education sector. The adoption of digital learning solutions can be facilitated, and technology competency can be validated with trustworthiness. To preserve confidence in the schooling industry, nevertheless, it is imperative to guarantee ChatGPT's accuracy and dependability (Hyun Baek & Kim, 2023).

Study implications

The comprehensive examination of factors impacting ChatGPT adoption has culminated in empirical evidence that has significant significance for technology developers, academia and industry. These ramifications take various forms, providing insightful information and practical solutions to promote the adoption, use and advancement of GAI platforms such as ChatGPT in Pakistani HE. This study highlights significant implications for the Pakistani education system. First, it highlights the importance of encouraging students and stakeholders to have a ChatGPT self-efficacious mindset. Educational institutions might potentially promote generative AI adoption and its efficient use for information transmission and engagement by offering resources, training and assistance to improve people's confidence in using technologies (Sahari et al., 2023). Second, the study emphasises how important trust is as a driver of new technology like ChatGPT adoption. The degree to which students are prepared to use ChatGPT for educational purposes is significantly influenced by their trust in technology providers and systems. Thus, governments and educational institutions must build trust in digital platforms by implementing reliable and transparent data security protocols, strong privacy safeguards and efficient communication tactics. Third, this study also highlights how crucial security measures are to encouraging the adoption of GAI technologies. Learners feel more confident using ChatGPT tools in a secure digital environment. Therefore, educational institutes must invest resources in creating strong policies, procedures and frameworks that improve accessibility. Fourth, using the TAM and its constructs in the setting of university students is a useful way to help future researchers believe that the study is real and credible. Notably, ease of use and utility significantly influenced students' behaviours towards ChatGPT. This research makes a substantial contribution to the advancement of conceptual models created for different OpenAI applications by offering substantial validation of parameters in the context of this novel technology. Lastly, teachers can assist in this area by providing reasonable norms and recommendations on using AI tools ethically and properly during assessments. Instructors can help students use ChatGPT effectively in assessments to ensure they understand the expected goal and its constraints.

Limitations and future direction

Even though this study has achieved significant advances, it is crucial to recognise its limitations because they open the door to more research in this field. First, our study is focused on the Pakistani university system, so the generalisability of our findings may be limited. For future studies, they could investigate the ways in which distinct societal, economic and regulatory issues impact different countries and areas. Second, cross-sectional data forms the basis of our investigation, which was carried out in a certain time period. Longitudinal research designs may be used in future investigations, providing stronger evidence of causation and alterations in the connection throughout time. Third, the self-reported participant data used in this study could be biased due to characteristics like the bias towards social desirability. Future studies could include objectives and factors of ChatGPT usage and self-efficacy to lessen the possibility of response bias. Fourth, our study focused on these factors since trust and self-efficacy are critical variables in ChatGPT adoption. Future studies should consider different factors such as perceived intelligence, awareness and cost to expand generalisability. Last, ease of use—the study's intermediate variable—might not adequately represent the shades of online debates and recommendations. Furthermore, future research should adjust to the rapidly changing technological environment to determine the application and significance of perceived security and confidence in emerging educational technologies.

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CONFLICT OF INTEREST STATEMENT

The authors declare no competing interests.

DATA AVAILABILITY STATEMENT

Data will be made available on request.

ETHICS STATEMENT

This research was conducted according to the Declaration of Helsinki guidelines and approved by Beijing University of Technology Ethics Committee. The patients/participants provided their written informed consent to participate in the study. No potentially identifiable human images or data are presented.

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